LESSON 10.1

Math is Everywhere

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find applications of math outside of school — in professions like being a chef, in hobbies like gardening, and around my own house.

WHY THIS MATTERS Some people think math is just a school subject with little relevance outside the classroom. But math is everywhere — chefs use it all day in restaurant kitchens, gardeners use it in planter beds, and it is hiding in jobs and hobbies all around you. A chef scales a recipe up for forty covers and works out how much stock to order. A gardener measures a planter bed and calculates the soil needed to fill it. Today you will name the math hiding inside jobs like these, and you will do some of it yourself — area, volume, rates, and estimates that have to be close enough to act on.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
profession *Spanish: profesión*	A job someone is trained to do, like chef, nurse, or carpenter.	A chef prices a menu; a nurse measures a dose; a carpenter cuts to length.
inventory *Spanish: inventario*	The supply of items a business keeps on hand, tracked by counting.	The kitchen counts 40 lb of flour and 12 dozen eggs on hand.
portion *Spanish: porción*	The measured amount of food served to one person.	Each bowl of soup uses 2 cups of broth — one person's serving.
area *Spanish: área*	The amount of flat space a surface covers, measured in square units.	A 4 ft by 2 ft planter covers 8 square feet of ground.
volume *Spanish: volumen*	The amount of space something fills, measured in cubic units.	4 ft × 2 ft × 1 ft of soil = 8 cubic feet.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

When chefs plan menus, they determine the number of items on the menu and calculate the price point for each item.

1. They decide on the portion size of each item and project how much food is needed for each day.
2. They calculate how much of each ingredient to order — that is counting, measuring, and multiplying, all before a single dish is cooked.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A planter that is 4 feet long and 2 feet wide has an area of 4 × 2 = 8 square feet — that tells us how many plants fit.

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The _____ needs _____ because the job is about _____.
- A chef uses math when they _____ .
- The math inside it is _____ .
- At home I used math when I _____ .

Word bank profession | inventory | portion | area | volume

WATCH OUT!

A common mistake

Believing math only 'counts' when it happens in a classroom

LESSON 10.1

Math is Everywhere

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

A planter is 4 feet long and 2 feet wide. What is its area?

(A) 2 square feet
(B) 6 square feet
(C) 8 square feet
(D) 12 square feet

HINT Area tells how much flat space the planter covers.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

That same planter is 4 feet long, 2 feet wide, and needs soil 1 foot deep. How much soil does the gardener need?

- (A) 4 cubic feet
- (B) 7 cubic feet
- (C) 8 cubic feet
- (D) 8 square feet

HINT Soil fills space, so you need volume.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Leaving the water running while brushing wastes an average of 4 gallons per brushing. You brush twice a day. How much water can one person save per day by only turning the water on to rinse?

- (A) 2 gallons
- (B) 4 gallons
- (C) 6 gallons
- (D) 8 gallons

HINT How many gallons does one brushing save?

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

Which of these is NOT one of the ways chefs use math when planning menus?

(A) Calculating the price point for each item

(B) Deciding on the portion size of each item

(C) Projecting how much food is needed for each day

(D) Choosing the color of the restaurant's walls

HINT Three of these came straight from the list of menu-planning math.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Name one task a chef does that uses math, and say what math is inside it.

SENTENCE STARTER A chef uses math when they ___ .
The math inside it is ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Name one way you used math around your house this week.

SENTENCE STARTER At home I used math when I ___ .
It was math because I had to ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you and your partner assigned to different workers. Which measurement does that job actually need — area, volume, or a portion — and how do you know from the task itself?

SENTENCE STARTER The ___ needs ___ because the job is about ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.1

Math is Everywhere

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

One person saves 8 gallons of water a day by turning the water on only to rinse. How much water does that person save in a year (365 days)?

(A) 365 gallons
(B) 373 gallons
(C) 1,460 gallons
(D) 2,920 gallons

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

If each person in a house saves 2,920 gallons a year, how much does a household of 3 people save in a year?

(A) 973 gallons
(B) 2,923 gallons
(C) 5,840 gallons
(D) 8,760 gallons

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

A household of 5 saves 2,920 gallons per person per year. A household of 4 does the same. How much MORE does the household of 5 save than the household of 4?

(A) 1 gallon
(B) 2,920 gallons
(C) 11,680 gallons
(D) 14,600 gallons

PER 1

4 Circle the letter of the best answer. Show how you know.

A gardener's planter has an area of 8 square feet, and each plant needs 2 square feet of space to grow. How many plants fit in the planter?

(A) 4 plants
(B) 6 plants
(C) 10 plants
(D) 16 plants

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 2 Explain and correct

reasoning

5 Answer in complete sentences.

Choose another person who works in a restaurant — not the chef — and describe the math they would use.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

What factors do you consider to determine the number of plants you can put in a planter garden? Is the TYPE of plant a factor? Explain your thinking.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you and your partner assigned to different workers. Which measurement does that job actually need — area, volume, or a portion — and how do you know from the task itself?

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LESSON 10.1

Math is Everywhere

Challenge · Extension

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- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each person to the math their real-world task uses.

_____ Chef tripling a recipe	(A) Multiply every ingredient by 3
_____ Gardener filling a 4 × 2 × 1 ft planter	(B) Volume: 8 cubic feet of soil
_____ Family tracking water savings	(C) 8 gallons × 365 days
_____ Shopper comparing two sizes	(D) Unit rate: price ÷ quantity

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Each person saves 2,920 gallons a year. Which correctly compares households of 3, 4, 5, and 8 people?

(A) 2,923 · 2,924 · 2,925 · 2,928 gallons

(B) 8,760 · 11,680 · 14,600 · 23,360 gallons

(C) 8,760 · 11,680 · 14,600 · 20,440 gallons

(D) 8,760 · 11,680 · 14,600 · 46,720 gallons

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A chef expects 120 customers tonight and projects that 1 out of every 4 will order the soup. Each serving of soup uses 2 cups of broth. How much broth should the chef prepare?

(A) 15 cups
(B) 30 cups
(C) 60 cups
(D) 240 cups

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Someone says, 'Math is just a school subject. I'll never use it after class.' Using the chef, the gardener, or the water problem, write what you would say back.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

Pick a hobby — yours or anyone's — and list at least three specific math tasks hiding inside it, the way the lesson did for gardening.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you and your partner assigned to different workers. Which measurement does that job actually need — area, volume, or a portion — and how do you know from the task itself?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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